

<Form 1> Project Summary ※ Must not exceed one A4 page.

Project Summary

Team name	SENTRY	
Project title		
Role	Name	Affiliation
Participants	Kim Changhoon Low-Cost Bridge Vibration Anomaly Detection System Robust to Temperature [name withheld] and Humidity Changes	CheongShim International Academy CheongShim International Academy
Advisor	[name withheld]	CheongShim International Academy
Project summary		
1. Project topic We developed a low-cost first-stage bridge vibration screening system that observes bridge vibration together with environmental variables using a low-cost MEMS accelerometer and a temperature/humidity sensor, compensates for normal variation explained by temperature and humidity, and detects only the remaining change as anomaly candidates. 2. Description of the project (proposed solution and its rationale) Damage in aging bridges accumulates between inspections, yet small and medium bridges lack a low-cost means of continuous monitoring. This project collects vibration and temperature/humidity with an ADXL355Z accelerometer, a DHT22 temperature/humidity sensor and a Raspberry Pi, compensates the frequency features with a temperature/humidity regression learned on normal data, and then trains an Autoencoder unsupervised on those residuals alone; when the reconstruction error exceeds the threshold (the normal 99th percentile), it flags an anomaly candidate and shows the bands that departed. The design rests on two facts: natural frequencies change with temperature and humidity even without damage, and damage data are hard to obtain from real bridges. 3. Differences from existing work (solutions) Existing low-cost MEMS SHM studies do not address environmental compensation, fiber-optic (BOCDA) precision measurement requires dedicated infrastructure, and unsupervised studies on public data mostly stop at outputting an anomaly score. This project is distinguished by integrating measurement, environmental compensation, anomaly judgment and band-wise explanation into a single low-cost device. 4. Results In the model-bridge experiment, adding 100 g on the left and right lowered the first peak frequency by 8.9%, confirming detection of mass change, and temperature/humidity compensation improved the F1 score about 4.9-fold, from 0.075 to 0.370, and the false-alarm rate in the humidity segment from 17.5% to 9.1%. On full-scale bridge data from Germany's openLAB, the false-alarm rate also decreased consistently after compensation.		